6. Reward Poisoning of Online-Learning Microarchitectural Policies

Motivation

Online reinforcement learning has moved from proposal to standard technique in microarchitecture. Pythia (MICRO 2021) learns prefetch decisions online from a reward that explicitly evaluates prefetch quality under current memory bandwidth usage. Micro-Armed Bandit (MICRO 2023) generalizes lightweight reusable RL to microarchitectural decision-making. Athena (HPCA 2026) arbitrates between prefetching and off-chip prediction with online RL. RL-CoPref and adaptive aggressiveness controllers extend the pattern.

Every one of these shares a property an adversarial-ML researcher would recognize immediately: the reward signal is computed from shared, adversary-influenceable system state. Pythia's reward depends on memory bandwidth usage — a resource any co-running process can modulate at will. The learner has no authentication of its reward, no anomaly detection, and no notion that its environment might contain an adversary.

Reward poisoning is a mature subfield in machine learning, with established results on stealthy reward perturbation, provable attack budgets, black-box poisoning of unknown learners, and characterizations of when a target policy can be induced within a bounded budget. The techniques exist and are well understood. They have never been applied to hardware learners, and hardware learners are in some respects easier targets: the state and action spaces are small, the reward function is published in the paper, the learning rate is fixed and documented, and the adversary shares the physical resource that generates the reward.

The consequences are more than denial of service. An adversary who can steer a victim's prefetcher into a chosen policy controls the victim's memory access pattern to some degree — a precondition for other attacks, and a route to amplifying a weak channel by forcing the victim into a configuration that leaks more. That is the framing that lifts this above a performance-degradation result.

Research gap

No published work examines adversarial manipulation of online-learning microarchitectural policies. The RL microarchitecture literature evaluates against benign co-runners; the adversarial RL literature targets software agents in simulated environments. The intersection is empty, and it is a genuine intersection rather than a mechanical transplant, because hardware learners have properties — bounded tables, quantized rewards, hard real-time inference, no software visibility — that change which attacks are feasible and which defenses are affordable.

Novelty

Closest prior work	What it actually did	Residual gap this proposal occupies
Bera et al., Pythia (MICRO 2021)	Online RL prefetching; reward evaluates prefetch quality under current memory bandwidth usage; reward levels exposed as configurable design knobs.	No adversarial model. The bandwidth-dependent reward is the attack surface, and the configurability provided for tuning is equally available for characterizing an attack.
Micro-Armed Bandit (MICRO 2023); Athena (HPCA 2026); RL-CoPref (2024)	Lightweight reusable RL for microarchitectural decisions; RL arbitration; coordinated prefetcher control.	All assume benign environments. Athena is a particularly interesting target because its arbiter chooses between mechanisms, so poisoning it steers which mechanism is active rather than merely degrading one.

Closest prior work	What it actually did	Residual gap this proposal occupies
Reward-poisoning literature (Ma et al. 2019; Rakhsha et al. 2020, 2021; Liu & Lai 2021; 2026 linear-MDP attackability characterization)	Reward, action and environment poisoning; stealthy perturbation; black-box attacks on unknown learners; necessary and sufficient conditions for attackability within a budget.	Entirely in software and simulated environments. No application to hardware learners, and no treatment of the constraints that make them distinctive — tiny state spaces, quantized rewards, no ability to intercept the reward directly, only to influence the resource it is computed from.
Cache and memory denial-of-service literature	Bandwidth and cache contention as availability attacks.	Degrades a victim by consuming resources. This proposal steers a victim's learned policy, which persists after the attacker stops and can target a specific policy rather than general slowdown — a qualitatively different capability.
Kosasih et al. (ASIACCS 2024)	Systematization of counter-based side-channel detection.	Relevant to whether poisoning is detectable, which should be evaluated, but does not address learned policies.

Novelty confidence is very high. Both literatures are mature and neither has crossed into the other. The main risk is that the crossing is judged mechanical, addressed by making the hardware-specific constraints central rather than incidental.

Proposed approach

Stage one: formalize the threat model. The adversary is a co-running process on another core. It cannot read or write the victim's RL state and cannot intercept the reward. It can modulate shared resources — memory bandwidth, LLC occupancy, memory-controller queue depth — and thereby influence the reward the victim's learner computes. This is an indirect, black-box, budget-constrained poisoning setting, and mapping it onto the existing adversarial-RL formalism is the theoretical contribution.

Stage two: the attacks. Three, in increasing sophistication:

- Degradation. Modulate bandwidth to drive Pythia's reward negative for actions that are in fact useful, steering it toward under-prefetching. Measure victim slowdown per unit attacker effort, and — the interesting part — how long degradation persists after the attacker stops, since a poisoned Q-table does not recover instantly.
    - Targeted steering. Induce a specific chosen policy rather than general degradation, following the policy-teaching formulation. On Athena, steer the arbiter to select the mechanism the attacker prefers.
- Stealth. Characterize the minimum attacker bandwidth footprint achieving a given degradation, and whether it is distinguishable from a legitimate memory-intensive co-runner. This separates a practical attack from a resource-exhaustion result, and is where the adversarial-ML budget formalism does real work.

Stage three: robust reward design. Candidate defenses spanning cost and effectiveness: normalize the reward against system-wide bandwidth so absolute contention does not shift it; clip or median-filter reward updates

to bound any single observation's influence, borrowing directly from robust-statistics defenses in adversarial RL; detect anomalous reward distributions and freeze learning; and partition learner state by security domain, expensive but structural. For each, quantify both residual attack success and performance cost on benign workloads — because a robust reward costing 3% of Pythia's benefit is a real trade an architect must weigh, and presenting it as free would be dishonest.

Evaluation

- Victim slowdown per unit attacker bandwidth, and persistence after the attacker stops.
- Targeted steering success rate: fraction of trials inducing the chosen policy, on Pythia and on Athena's arbiter.
- Stealth: attacker footprint versus effect, and distinguishability from a benign memory-intensive co-runner using performance counters.
- Defense: residual attack success and benign-workload performance cost for each candidate.
- Generality: repeat against Micro-Armed Bandit to show the vulnerability is a property of the online-RL pattern rather than of Pythia specifically. This makes the paper about a class rather than a design.
- Sensitivity to learning rate and reward quantization, since these are parameters a defender can adjust cheaply.

Work plan

Phase	Principal activities	Deliverable
1	Integrate the Pythia artifact and an adversarial co-runner model in ChampSim; reproduce published Pythia performance.	Validated baseline
2	Formalize the indirect black-box poisoning threat model; map onto the adversarial-RL formalism.	Threat model — theoretical contribution
3	Build degradation and targeted-steering attacks; measure persistence.	Attack results
4	Stealth characterization; detectability study.	Practicality result
5	Design and evaluate robust reward mechanisms; generality against Micro-Armed Bandit and Athena.	Defenses + generality
6	Writing + artifact.	Submission + artifact

Key references

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